MIL-W-17265F(SHIPS)
4 January 1974
SUPERSEDING
MIL-W-17265E(SHIPS)
21 February 1967
(See 6.2)

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED (ELECTRIC-HYDRAULIC, VARIABLE SPEED);

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for the design and construction of electric-hydraulic winches for shipboard applications. The intent is to set forth the ambient conditions within which the equipment must operate satisfactorily and reliably, to specify general design and test requirements. Throughout the design and manufacture of the equipment, maximum effort shall be made to attain basic design objectives to insure that the equipment will meet the needs of the Naval service (see 3.3). Requirements applicable to specific winches shall be as specified in the individual equipment specification.

1.2 Classification. Electric-hydraulic winches shall be furnished in the types specified in the applicable detail specification (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-85 - Screw, Cap, Slotted and Hexagon Head.
FF-S-86 - Screw, Cap, Socket Head.
FF-S-92 - Screws, Machine: Slotted, Cross-Recessed or Hexagon Head.
FF-S-107 - Screws, Tapping and Drive.
FF-S-200 - Set Screws, Hexagon, Socket and Spline Socket, Headless.
FF-S-210 - Set Screws, Square Head and Slotted, Headless.
HH-L-361 - Linings, Brake Automotive Use.
QQ-I-652 - Iron Castings, Gray.
QQ-P-416 - Plating, Cadmium Electrodeposited.
TT-E-490 - Enamel, Silicone Alkyd Copolymer, Semigloss, Exterior.
TT-P-57 - Paint, Zinc, Yellow-Iron Oxide-Base, Ready-Mixed.

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification For.
MIL-S-61 - Shunts, Instrument, External, 50 Millivolt (Lightweight Type).
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-713 - Twine, Impregnated, Lacing and Tying.
MIL-B-857 - Bolt, Nuts and Studs.
MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-P-1144 - Pipe, Stainless Steel (Corrosion-Resistant), Seamless or Welded.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-P-3184 - Packaging of Machinery: Deck and Vehicle Mounted with Associated Equipment and Repair Parts.
MIL-I-3190 - Insulation Sleeving, Electrical, Flexible, Coated, General Specification For.
MIL-S-3786 - Switches, Rotary (Circuit Selector, Low Current Capacity) General Specification For.
MIL-S-3950 - Switch, Toggle, General Specification For.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-F-5591 - Fasteners; Panel.

FSC 3950

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MIL-R-5757 - Relays, Electrical (For Electronic and Communication Type Equipment), General Specification For.
MIL-R-6106 - Relay, Electric, Aerospace, General Specification For.
MIL-I-7444 - Insulation Sleeving, Electrical, Flexible.
MIL-T-7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification For.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push, (Snap Action), General Specification For.
MIL-S-8834 - Switches, Toggle, Positive Break, Aircraft, General Specification For.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification For.
MIL-F-15160 - Fuses, Instrument, Power and Telephone.
MIL-S-15291 - Switches, Rotary, Snap Action.
MIL-E-16366 - Electrical Clamps, Lug Terminals and Conductor Splices - Pressure Grip.
MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded.
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature.
MIL-I-16923 - Insulating Compound, Electrical, Embedding.
MIL-M-17060 - Motors, 60 Hertz, Alternating Current Integral Horsepower (Shipboard Use).
MIL-I-17244 - Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 Inch Dial).
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure, Shipboard Use.
MIL-C-17588 - Circuit Breakers (Automatic - ALB) and Switch, Toggle (Circuit Breaker, Non-Automatic - NLB), Air, Insulated Enclosure, 125 Volts and Below, AC or DC, Naval Shipboard.
MIL-P-17869 - Pumps and Motors, Power, Oil Hydraulic (Naval Shipboard Use).
MIL-S-19500 - Semiconductor Devices, General Specification For.
MIL-R-19648 - Relays, Thermal, Time Delay, Hermetically Sealed, General Specification For.
MIL-S-21604 - Switches, Rotary, Multipole and Selector Type, 1 to 10 Amperes.
MIL-S-22885 - Switch, Push Button, Illuminated, General Specification For.
MIL-C-22992 - Connectors, Plugs and Receptacle; Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specification For.
MIL-I-23053 - Insulation Sleeving, Electrical, Heat Shrinkable, General Specification For.
MIL-C-23742 - Control Centers, Alternating Current, Naval Shipboard.
MIL-H-24135 - Hose, Reinforced, Water and Oil Resistant, and End Fittings, Reusable, For Flexible Hose Connections.
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) for Shipboard Application.
MIL-F-24402 - Filter and Filter Elements, 25 Micrometre Absolute, Ship Hydraulic Systems.
MIL-H-24459 - Hydraulic Fluid, Antiwear Petroleum Base.
MIL-N-25027 - Nut, Self Locking; 250°F, 450°F, 500°F and 800°F; 125 KSI, FTU; 60 KSI, FTU and 30 KSI, FTU.
MIL-C-26482 - Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting, General Specification For.
MIL-M-28787 - Modules, Electronic, Standard Hardware Program, General Specification For.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-242 - Electronic Equipment Parts Selected Standards.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-681 - Identification Coding and Application of Hook up Wire.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-1378 - Requirements for Employing Standard Hardware Program Modules.
MIL-STD-1389 - Design Requirements for Standard Hardware Electronic Modules.
MIL-STD-1399, Section 103 - Interface Standard for Shipboard Systems - Electric Power, Alternating Current.

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MILITARY (continued)
MIL-STD-1472 - Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MS15004 - Fittings, Lubrication (Hydraulic) Surface Check, 1/4-28 Taper Threads, Nickel-Copper Alloy, Type IV.
MS15005 - Fittings, Lubrication, Throat or Surface Check, 1/8 Pipe Threads, Nickel-Copper Alloy, Type V.
MS27212 - Terminal Board Assembly, Molded-In Stud, Electric.
MS33540 - Safety Wiring and Cotte. Pinning, General Practices For.
MS33649 - Boss, Fluid Connection-Internal Straight Thread.

AIR FORCE-NAVY AERONAUTICAL STANDARD
AN 814 - Plug and Bleeder - Screw Thread.

HANDBOOKS

MILITARY
MIL-HDBK-216 - RF Transmission Lines and Fittings.
NAVWEPS OD 30355 - Standard Hardware Program.
H28 - Screw Thread Standards for Federal Services.

DRAWING

MILITARY
S2601-860303 - Capstan and Gypsy Heads, Tabulation of Sizes.

PUBLICATION

MILITARY
NAVSHIPS 0901-000-0010 - Preservation of Ships Service - Volume I.

See supplement-1 for list of applicable detail specifications.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A106-72 - Seamless Carbon Steel Pipe for High-Temperature Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
B16.5-68 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)
Surface Preparation Specification No. 10 - Near White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 5th Avenue, Pittsburgh, Pennsylvania 15213.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
ICS-70 - Industrial Controls and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern.

3.2 Material. Materials for mechanical items shall be in accordance with either commercial or Government specifications unless otherwise specified in the contract or order. If commercial materials are used the bill of material shall list both the commercial specification and the equivalent Government specification. The use of cast iron is limited to material conforming to MIL-I-24137 or QQ-I-652, class 35 or above. Radiographing of steel castings is not required.

3.3 Design. The winches shall be designed to satisfy all the requirements specified herein and to demonstrate compliance by passing the tests specified herein. The design philosophy for the electric-hydraulic winches is to utilize the latest construction techniques, to provide the reliability and maintenance requirements specified herein. Manufacturers are encouraged to forward to NAVSEC proposals and suggestions to further promote this philosophy.

3.3.1 Environment. Winches shall be designed for continual exposure to winds up to 40 knots, salt spray environment, and temperatures that may vary from extremes of plus 10°F to plus 120°F.

3.3.2 Operating life. The design life of the winch system shall be 10,000 operating hours. The winch system shall be designed for 1000 hours operation between overhauls without ajustments or parts replacement.

3.3.3 Human engineering. The supplier shall design the equipment using the human engineering design criteria principles and practices contained in MIL-STD-1472.

3.3.4 Maintainability program. The supplier shall provide and maintain a maintainability program in general accordance with MIL-STD-470.

3.3.4.1 Program requirements. The maintainability program shall include at least, but not limited to the following items:

(a) Maintainability management and control plan. This plan will indicate the position of the management within the overall company organizational structure.
(b) Maintainability prediction. A maintainability prediction will be done according to MIL-STD-470.
(c) Task and skill analysis. A task and skill analysis will be done to determine the level and extent of the capability that is required to perform maintenance actions.
(d) Maintainability demonstration testing. Testing will be undertaken to determine the validity of the prediction and established estimates of the mean-time-to-repair (MTTR) the system. Maintainability testing shall be accomplished using MIL-STD-471 as a guide. Test plan shall be submitted to the Naval Ship Engineering Center (NAVSEC) for review prior to initiation of tests.
(e) Maintenance concept. See 3.29.2(c) for maintenance concept.

3.3.4.2 Maintainability criteria. As a minimum, the maintainability criteria specified in 3.3.4.2.1 through 3.3.4.2.4 shall be used.

3.3.4.2.1 Operational preventive maintenance services. Minimum time between preventive maintenance (PM) services shall be 5 days or 50 hours of operation, whichever occurs first.

3.3.4.2.2 Disuse preventive maintenance services. PM services during periods of disuse shall guard against degradation of the winch for periods up to 120 days. During disuse the winch may be exercised at a frequency not greater than once every 30 days. Exercise requirements shall be limited to 1 hour duration and shall not require the use of special auxiliary machinery or test loads.

3.3.4.2.3 Maximum allowable time to repair. Maximum allowable time to repair, based upon replacement of subassemblies, shall be 12 hours.

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3.3.4.2.4 Mean allowable time to repair. Mean allowable time to repair, based upon replacement of subassemblies, shall be 3 hours.

3.4 General requirements. The winch shall consist of drums and gypsy heads, as specified herein, mounted horizontally on the same shaft, with necessary reduction gearing, electric motor, control panel, hydraulic pump and motor, hydraulic brake, local control, remote control stand complete with controls, and all operating accessories necessary for operation, exlcusive of power leads. The winch shall be self contained on a single rigid bedplate except for the remote control stand and control panel which shall be suitable for remote installation. The arrangement of all winch components on the bedplate with respect to each other shall be the same for all winches. The individual part requirements shall be as specified herein and in accordance with the applicable detail specification.

3.4.1 Control lever and hand wheel location. Drum pawl and ratchet, drum brake, drum clutch, and hydraulic brake operating and release mechanisms shall be centrally located at the end of the winch, opposite the wire rope lead and clear of gypsy head rope leads.

3.5 Drums. The drums shall have barrel diameters, widths and storage capacities as specified herein. Each drum shall have adequate flange depth to stow the specified length and size wire rope with a minimum of 2-1/2 inches of clear flange above the top layer. Each drum shall be guarded as necessary to prevent the wire rope from jumping the drum.

3.5.1 Stowage capacity and average layer. Stowage capacity and average layer of the drum shall be determined by the following formulas:

$L = (A + B) \times A \times C \times F$
$D = B + A + d/2$
L = Capacity of drum in feet of wire
A = Depth of wire spooled on drum
B = Barrel diameter
C = Drum width
D = Diameter of average layer
d = Wire diameter
F = Spooling factor = $\frac{.262}{d^2}$ (controlled winding)
$\frac{.210}{d^2}$ (uncontrolled winding)

3.5.2 Wire rope drum attachment. All winch drums shall be provided with a wire rope dead end anchor device designed to:

(a) Withstand a line pull equal to twice the rated load with two or more turns of wire rope on the bottom layer.
(b) Release the bitter end of the wire rope at a tension of not more than 2,000 pounds when the last two turns are unspooled. The bitter end shall be released in a manner that will let it run free through the rigging.

3.6 Drum clutch. Each drum shall be equipped with a positive type clutch capable of being engaged in less than 90 degrees of drum rotation. The clutch yoke lever shall extend beyond the movable clutch section and shall be equipped with a positive means of locking the clutch in both the engaged and disengaged positions. The clutch shall be visible to the operator so he can determine by visual examination when the clutch sections are in alinement and when they are engaged. The clutch lever shall be rigid enough to prevent an operator from locking the handle in the engaged position without the clutch actually being engaged.

3.7 Brake. Each drum shall be equipped with an externally contracting flat bank brake operated by a handwheel (minimum diameter of the handwheel shall be not less than 12 inches). The drum brake shall be designed to hold a static load of 150 percent of the winch rated line pull, with the wire rope on the top layer. The effort to set the brake to its specified static capacity shall not exceed a total of 60 pounds applied on the handwheel. The brake mechanism shall be designed to withstand a force of 200 pounds on the handwheel without creating excessive loads on the drum shaft and drum shaft bearings (supporting calculations shall be included with the stress calculations). The brake lining shall be in accordance with HH-L-361 and the unit pressure shall not exceed that recommended by the manufacturer for the particular application. Positive means shall be provided for holding the brake band clear of the drum when the brake is released. The brake drum and brake band shall be designed to afford maximum protection to personnel and to preclude the entrance of foreign matter under the brake band. Means shall be provided to permit inspection, repairs and adjustment for wear.

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3.8 Ratchet and pawl. Each drum shall be equipped with a ratchet and pawl designed to hold the drum under a line pull equal to the breaking strength of the wire rope specified with the wire rope on the top layer. The pawl shall be heavily counter-balanced or spring controlled so as to fall out of engagement when the load on the pawl is relieved. This lever shall also be fitted with a locking pin, ratchet, or similar means which will positively secure the pawl in the "out" position.

3.9 Gypsy heads. All winches shall be designed for two gypsy heads. The design of the winch shall provide clear leads both fore and aft at each gypsy head. Each gypsy head shall be surfaced hardened and proportioned in accordance with Drawing S2601-860303 for five inch circumference manila rope, maximum breaking strength 22,500 pounds. In the event less than two gypsy heads are specified to be supplied with the winch, a shaft cap shall be provided for the ends of the drum shaft not provided with a gypsy head. The design of the gypsy head and shaft cap shall be such that either may be installed on the drum shaft without the use of special tools. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.10 Gearing. Gear drives shall be used throughout. All gearing shall be in accordance with American Gear Manufacturer's Association (AGMA) publications. All gearing shall be single or double helical type and shall have machine cut teeth. All gearing shall be totally enclosed in an oil tight case and shall be lubricated by an oil bath. Each gear case shall be fitted with a drain which will permit complete draining of the gear case.

3.10.1 Holes for gearing. All housings containing gears shall be provided with oil tight covered openings in sufficient number to allow visual examination of the teeth on all gears. Such openings shall expose to view at least two gear teeth at a time. The openings shall permit viewing the mesh between gears, if practicable.

3.10.2 Dipstick. A dipstick shall be provided for indicating the oil level in each gear case. The dipstick should be marked to indicate the "FULL" and "ADD OIL" oil levels.

3.11 Roller or ball bearings shall be used for gear and drum shafts and all accessories and accessory mechanisms, such as drum pawls, clutches, brakes, and operating mechanisms shall use needle or roller bearings. In all applications of anti-friction bearings, the assembly drawing of the winch shall bear a certificate signed by a responsible official of the bearing manufacturing company showing that with full consideration to the intended service requirements, the bearing applications proposed are approved by his company as to all details of mounting, loading, speed and lubrication.

3.12 Bedplate. The bedplate shall be of adequate strength to maintain alinement without aid from the deck foundation. The thickness of the bedplate flanges shall be at least equal the diameter of the hold down bolts used. Foundation bolts holes shall be drilled 1/16 inch undersize for reaming at installation to fit body bound bolts. The design shall eliminate pockets in which water can stand or collect. For design of controls, the top of the foundation may be assumed to be 6 inches above the deck.

3.13 Fasteners. Protective means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. All bolts, cap screws and studs shall be in accordance with MIL-B-857, FF-S-85, FF-S-86 or FF-S-92. Fasteners with a nominal diameter of 1/2 inch or less and exposed with the unit in operating position shall be of corrosion-resistant material. Set screws, if used, shall have positive locking devices. Screw threads shall be unified, coarse, or fine thread series, in accordance with Handbook H28.

3.14 Guards. All rotating shafts shall be provided with protective covers or guards. Other guards shall be provided as necessary to protect winch components from being damaged by slack wire dragging across the winch.

3.15 Lubrication. All bearings shall be provided with pressure grease fittings. Sliding clutches, unless operating in an oil bath, shall be provided with at least two grease fittings, 180 degrees apart. All grease fittings shall be in accordance with MS15004 or MS15005. All grease fittings shall be recessed or otherwise protected to preclude them from being damaged. Individual grease fittings and passage ways shall be provided for each point to be lubricated.

3.16 Hydraulic equipment.

3.16.1 Hydraulic transmission. The hydraulic transmission for the winch shall be designed as a self contained package unit. The package shall consist of hydraulic pump and motor, appropriate valving, filters, controls, vents and hydraulic sump tank and shall be

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designed so that it can be installed on and removed from the winch bedplate, without making or breaking any hydraulic connections between pump, motor, valves, filters and controls. To reduce to a minimum, the amount of hydraulic tubing and fittings used, all valves, filters and gage connections which are not integral to the hydraulic pump and motor cases shall be, where practical, mounted on a single steel manifold. Main pumps and motor shall be mounted to the manifold so that main loop hydraulic system runs are incorporated within the manifold. External lines from the hydraulic transmission to hydraulic brakes, air-oil coolers, and anti-slack devices (where specified) shall be provided with self-sealing fittings to permit making or breaking of hydraulic connections. All hydraulic components shall be mounted within a single housing with only the hydraulic pump and motor shafts and specified controls external from the housing.

3.16.1.1 The hydraulic transmission package shall be designed for 1000 hours of operation or 3 years of service (whichever occurs first) without failure of any components. To insure that the cleanliness of the system is not compromised, seals shall be applied prior to shipment. The seals shall be installed in such a manner that a broken seal would indicate that the cleanliness of the hydraulic system had been compromised.

3.16.2 Hydraulic pumps and motors. Hydraulic pumps and motors shall be in accordance with MIL-P-17869, class I or II, except that the differential pressure (p) for rated line pull and line speed of the winch shall not exceed 2000 pounds per square inch (psi). The rotating speed of the main pumps and motors shall not exceed the speed at which the pump and motor was qualified at for any operating condition. A mechanical position indicator shall be provided to indicate the setting of the hydraulic pump. If a variable delivery type of hydraulic motor is used, it shall be provided with a mechanical position indicator to show the setting of the motor. The mechanical position indicator shall be mounted on the outside of the pump motor housing.

3.16.2.1 Piping or tubing used inside the hydraulic sump tank shall be limited to the following:

(a) Seamless carbon steel pipe in accordance with ASTM A106-72.
(b) Corrosion-resisting steel pipe in accordance with MIL-P-1144.
(c) Copper-nickel (70-30) tube in accordance with MIL-T-16420.

3.16.2.2 Piping or tubing used outside the hydraulic sump tank shall be limited to copper-nickel (70-30) tube in accordance with MIL-T-16420.

3.16.2.3 Flexible hose conforming to MIL-H-24135, class 3 may be used outside the hydraulic sump tank.

3.16.2.4 Tapered pipe thread connections in the piping system are prohibited. Take-down pipe connections shall be ANSI B16.5-60 flanges modified for O-ring face seals or Combination Pump Valve (CPV) Company O-ring face seal unions, or equivalent. Connections to components may be straight thread O-ring fittings. All penetrations into the primary hydraulic systems for instruments and test points shall be MS33649 straight thread O-ring bosses. All test points and ports not having permanently installed instruments shall be closed with AN 814 safety plugs.

3.16.3 Pressure compensator and relief valve. The hydraulic circuit shall be provided with a pressure compensator and adjustable relief valves. The pressure compensator shall be designed to reduce the stroke of the pump to zero when the hydraulic pressure reaches 165 percent of the rated operating pressure. The main system relief valves shall be set above the pressure compensator pressure setting but not in excess of 180 percent of the rated operated pressure. The relief valves shall have a volumetric capacity of not less than the full displacement of the hydraulic pump.

3.16.4 Hydraulic sump tanks. Each hydraulic sump tank shall be designed so that it will not breathe atmospheric air either directly or indirectly. The tanks shall be designed so that the hydraulic fluid can be drained from the tank without spilling on the deck.

3.16.4.1 The fill and drain connections for the hydraulic sump tank shall be remotely located from each other to ensure maximum circulation of hydraulic fluid in the sump when connected to a hydraulic purifier. Both the fill and drain lines shall maintain a minimum pipe fitting size of 1-1/4 inch NPT throughout. Both the fill and drain connections shall terminate in a 1-1/4 inch NPT male fitting. An end cap shall be provided for the end of each fill and drain connection. The drain line shall incorporate a hand operated shut-off valve. The fill line shall discharge approximately six inches below the normal fluid level in the sump tank.

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3.16.4.2 A circulating pump driven by its own electric motor shall be provided for the purpose of (1) providing hydraulic fluid to the servo pump, (2) providing make-up hydraulic fluid (replenishment) to the active system, (3) providing a means for heating the hydraulic fluid, (4) providing a means for continual filtration of the hydraulic fluid and (5) providing hydraulic fluid to the main pump and motor cases. The electric motor for the circulating pump shall run continuously except for periods of maintenance. The motor, motor controller and master switches for the circulating pump electric motor shall conform to 3.20 and 3.21. The circulating pump shall not be driven at a speed in excess of 1800 revolutions per minute (rpm).

3.16.4.3 The winch motor controller for the main hydraulic pump shall be interlocked to the replenishment system so that it will not be possible to start the winch motor without proper replenishment pressure.

3.16.5 Filters. The filter installed between the storage and reservoir as well as any others installed in vital points of the system shall conform to MIL-F-24402. Filters shall have disposable type elements. All filters shall be provided with pressure differential indicators. Filter cases shall have a built-in by-pass valve. All filters that are to be serviced by ships force shall be located so that filter elements may be replaced or inspected without opening the hydraulic sump tank.

3.16.6 Precharged accumulators. If precharged accumulators are used in any part of the hydraulic system each accumulator shall be provided with a pressure gage to monitor the charge and such valves and fittings as necessary to permit recharging of the accumulator. Pressure gages and fittings for recharging the accumulator shall be outside of the sump tank.

3.16.7 Air vents. Air vents shall be provided at all points in the hydraulic system where air will tend to accumulate and may result in unsatisfactory operation. Air vents shall be accessible after installation on the winch. Disconnection of pipe lines or loosening of flanges is not an acceptable way of venting air.

3.16.8 Horsepower limiter. The winch shall be provided with a horsepower limiter. The horsepower limiter shall limit the horsepower of the winch under any load to approximately the horsepower required to handle rated load at rated speed. The horsepower limiter shall not limit the payout speed.

3.16.9 Air-oil cooler. An air-oil cooler shall be provided as an integral part of the winch. Based on an ambient temperature of 90°F, and the duty specified, the air-oil cooler shall limit the maximum temperature of the hydraulic fluid in the sump tank to 150°F. All hydraulic fluid from the pump and motor case drains shall pass through the air-oil cooler.

3.16.9.1 The air-oil cooler shall consist of cooling coils, electric motor driven ventilation fan, valves, hydraulic lines and fittings for connection to the hydraulic transmission, necessary control components to automatically activate the cooler as required.

3.16.9.2 The speed of the hydraulic motor driving the ventilation fan shall be automatically controlled so that the fan speed is proportional to the cooling requirements.

3.16.10 Cold weather operation. The circulating system (see 3.16.4.1) shall be designed so that it will heat the hydraulic fluid as the fluid is being circulated. The circuit shall be capable of maintaining the hydraulic fluid at a minimum of 60°F when the ambient temperature is 10°F and the unit is exposed to winds up to 40 knots.

3.16.11 Cleaning. All hydraulic pumps, motors, valves, pipes, tubing, fittings and reservoirs shall be thoroughly cleaned prior to assembly to insure the contamination level does not exceed the hydraulic manufacturer's recommendations.

3.16.12 Hydraulic fluid. Hydraulic fluid for the hydraulic system shall conform to MIL-H-24459.

3.16.13 Fluid level. A sight gage shall be provided to indicate the level of fluid within the sump tank. The gage shall show the normal level of fluid as well as the high and low levels. The gage shall be designed so that it will not expose the system to contamination.

3.16.14 Indicator. An indicator conforming to MIL-I-17244 shall be installed in the hydraulic sump tank. The indicator shall be in a position easily monitored.

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3.16.15 Fluid velocity. Sizes of piping and passages shall be such that the velocity of fluid in pressure lines does not exceed 20 feet per second (fps) except when other considerations justify higher velocities. Suction lines shall be sufficiently large, taking bends and other restrictions into consideration, to insure sufficient pressure at pump intake ports at the minimum fluid temperatures. The maximum fluid velocity in pump suction lines having no positive head shall be limited to 4 fps.

3.16.16 Hydraulic transmission bedplate. The bedplate for the hydraulic transmission package shall be designed so that it can be readily installed on a removal from the winch. The hydraulic transmission package shall be of adequate strength to maintain internal alinement during handling and shipment when fully charged with hydraulic fluid.

3.16.17 Electrical control enclosure. A watertight enclosure shall be provided as a part of the factory sealed hydraulic transmission package. The enclosure shall provide for all external electrical connections, electrical controls, time recording devices, heater controls, etc. The enclosure shall conform to NEMA ICS-70.

3.16.18 Timer. An electrical timing device shall be provided which will record the number of hours the hydraulic transmission is operated. The timing device shall record up to 9,999 hours of operation. The timing device shall be mounted on the transmission in a watertight enclosure. Electrical leads to the timing device shall be installed so that they may be readily connected or disconnected in the event the transmission must be removed from the winch.

3.16.19 Hydraulic brake. The high speed pinion shaft shall be equipped with a hydraulically released spring setting externally contracting disk brake with an external manual release (hand lever) which will automatically reset the brake when the operator releases the hand lever. Effort to release the brake shall not exceed 60 pounds on the hand lever. The brake shall be designed to stop and hold 150 percent winch rated load when lowering at rated speed (wire rope on average layer) and to withstand a static pull of 200 percent of the rated load without creepage.

3.16.19.1 The hydraulic brake shall be designed so that the brake will release at not more than 4 percent pump stroke and shall set at not less than 3 percent stroke in either direction. The hydraulic brake control valve shall be operated direct or through a pilot valve as required to provide the accuracy specified herein. Cross venting or open center shall be provided between the high pressure and low pressure hydraulic line when the hydraulic brake is set. The hydraulic brake shall automatically set upon loss of servo pressure, replenishment pressure or electric power.

3.16.20 Instruction plate. An instruction plate shall be attached to the sump tank near the filling point and located so it can be easily read. The instruction plate shall contain complete instructions for adding fluid to the tank and shall in addition include the following information:

(a) Total fluid capacity of the system.
(b) Sump tank capacity (to normal level in the tank).
(c) Specification or stock number of hydraulic fluid to be used.

3.16.21 Sampling valve. A hydraulic fluid sampling valve shall be provided for taking samples from the high pressure line between the pump and motor. Sampling valves shall be protected against inadvertent damage by other equipment and backwired in accordance with MS33540. Sampling valves shall be Aircraft Porous Media, Inc. Part No. AC-7893-2186A, or equal. One number AN814-8L plug and bleeder in accordance with AN814, to be used when valves are serviced, shall be supplied with each valve.

3.17 Electrical/electronics - general requirements.

3.17.1 Electronic controls shall be designed using Navy Standard Hardware Program (SHP) modules in accordance with MIL-STD-1378 for all control circuitry within the constraints specified in MIL-STD-1378. Existing hardware designs are specified in volume II "Module Parameters" of OD 30355 and shall be used to the maximum extent possible. In the event appropriate SHP designs are unavailable, MIL-M-28787 and MIL-STD-1309 shall be used as the development document for new designs.

3.17.2 Power supply. The power input to the winch system shall be a single input from a 440 volt, 3 phase, 60 hertz (Hz), type I power source as defined in MIL-STD-1399, section 103. Control power shall be provided by control transformer and other conversion equipment as required and shall be supplied as part of the controller. Continuously operating space heaters may utilize 115 volt power supplied separately from the ship's lighting circuit.

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3.17.3 Circuit voltages. 440 volt, 3 phase, 60 Hz current shall be used only for power supply to electric motors. All other control, pushbutton and power circuit voltages shall not exceed 120 volts.

3.17.4 Through deck penetrations. The supplier shall cable all winch electric components to a winch mounted junction box so that shipyard furnished cables, which must penetrate the weather deck, are limited to the following runs:

(a) One cable conducting 115 volts alternating current (vac) power leads from the main winch to a below decks junction box furnished by the installing activity.
(b) One shielded cable conducting electronics leads from the main winch to a below decks junction box furnished by the installing activity.
(c) One cable as in (a) above from the master control console.
(d) One cable as in (b) above from the master control console.
(e) One 440 volt power cable from each electric motor to its motor controller.
(f) One cable from each constantly operated heater to a below decks distribution panel furnished by the installing activity.

3.17.4.1 No cables shall run directly between the winch and the master control console.

3.17.5 Safety. Equipment shall be designed and constructed to ensure safety to operating and maintenance personnel. When the equipment is properly installed and the enclosure is grounded, there shall be no accessible way for operating personnel to receive an electric shock even though an internal fault may exist between any two circuits, between any circuit and a structural member, or between any circuit and ground. Component design shall hold to a practical minimum the possibility of maintenance personnel being exposed to electric shock while servicing or checking out the equipment. External projections which are a potential hazard to personnel shall be avoided wherever practicable. When their use in unavoidable, positive protection in the form of a guard shall be provided. Sharp corners and projections which may cause injury in a rough sea or on which clothing may catch shall be avoided. Equipment shall be designed to revert to its least hazardous condition or mode of operation upon failure of a circuit or part. Special performance capabilities (continuation of operation, automatic shut-down or any other specific function) in the event of an electrical part failure shall be as required by the equipment specification. The design shall be such that should failure occur, any resulting damage will be confined to the smallest equipment subdivision (minimum replaceable part or subassembly) within which failure occurred.

3.17.6 Compatibility. Materials and parts used in the equipment shall be compatible with each other and with the environment for which they are intended. For example, silicone materials such as lubricants and insulation generally produce excessive brush wear in nonventilated motors and generators; therefore, such materials shall not be used in combination in such equipment. Compatibility shall be considered from the standpoint of economy as well as from the chemical, physical and electrical aspect. For example, inexpensive short-life parts shall not be combined with expensive long-life parts in a nonrepairable subassembly. If it is not practicable to make them individually replaceable, parts having similar reliability and life expectancies for conditions under which they are used shall be grouped together.

3.17.7 Economy. Equipment shall be designed for maximum overall economy taking into account procurement, operating, and maintenance costs for the specified life.

3.17.8 Accessibility. All parts which may require servicing, repair, or replacement during the life of the equipment shall be readily accessible for repair or replacement. Access to internal parts of equipment other than rotating machinery shall be from the front of the enclosure since units are likely to be installed with their backs to bulkheads and their sides adjacent to other equipment. Operating controls shall be conveniently located on the front of the enclosure or on an operating panel appropriately oriented with respect to the operator's station. Nonoperating controls shall be located behind access doors within the enclosure or in other accessible locations within the equipment. Adjustments such as for calibration, compensation, or alinement shall be located within the equipment and shall be readily accessible when the equipment is opened for servicing.

3.17.8.1 The main electronics cabinet shall be designed so that a minimum of three inches clearance is provided between the sides, bottom, and top of the enclosure and any internal component.

3.17.8.2 Terminal boards shall be located so all terminal lugs are readily accessible for meter probe readings without disturbing other components.

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3.17.9 Interchangeability. All parts, subassemblies and assemblies which are removable shall be interchangeable with corresponding items from a similar equipment, from stock or from production without the necessity for re-forming (for example, drillings, bending, filing) or the use of undue force and without the necessity for individual selection.

3.17.10 Simplicity of ease of operation. Simplicity and ease of operation shall be accomplished by keeping the number of controls to a minimum and by using a minimum but logical sequence of operating procedures.

3.18 Electrical/electronic controls.

3.18.1 Function. Electrical/electronic controls shall provide all electrical/electronic power required to safely operate, test, and maintain the type winch for which it was designed. The controls shall also provide manual and automatic controls with which to energize the electrical inputs required.

3.18.2 Test module. The electronic control box shall incorporate a SHIP test point module.

3.18.2.1 Test points. Test receptacles shall supply all pertinent test voltages, currents, and all other readouts necessary to:

(a) Safely check the entire winch system.
(b) Safely isolate the general area of malfunction, i.e., electrical/electronic, hydraulic, mechanical.
(c) Safely isolate the electrical/electronic malfunctioning module.
(d) Perform regular preventive maintenance programs.
(e) Safely operate the complete system on a regular basis.

3.18.2.2 Test data shall reflect a logical sequence of all operations.

3.18.2.3 Test data describing a logical sequence of all testing operations described in 3.18.2.1 shall be supplied as part of all maintenance manuals.

3.18.2.4 A test operated switch shall be provided in the electronic control box to facilitate testing without the electric motor running. The switch contacts shall be automatically opened when the electronics cabinet door is closed so that control of the electronics power is returned to the main motor starter relay contacts.

3.18.3 Primary power circuits. Primary power circuits shall not be directly grounded. Where capacitive type of grounding is necessary, such capacitance shall be as small as practicable. All leads from any primary power supply shall be individually protected against damaging overload between the service connection and any other part of the system, equipment, or unit, as applicable.

3.18.3.1 The electronic circuit shall be designed so that all circuits which conduct 115 vac power to weather-deck electrical components such as solenoid operated valves or selector switches shall be isolated from the electronic circuit board printed land patterns and components.

3.18.4 Circuit breakers. Circuit breakers for direct current (dc), single phase and three phase applications shall conform to the following:

(a) 0.050 to 15 amperes at 50 volts direct current (vdc) or 115 vac - MIL-STD-242.
(b) Greater than 15 amperes at voltages up to 115 vac or dc - MIL-C-17588.
(c) Greater than 15 amperes at 250 vdc or 500 vac - MIL-C-17361.

3.18.5 Power interruption. The equipment shall be capable of withstanding, without damage, the following conditions of the ship's power supply:

(a) An interruption with power restoration occurring anywhere from 3 to 30 seconds (during transfer from normal to alternate or emergency power supply).
(b) High voltage transients of three and one-half times nominal voltage but of short duration (less than 1 cycle).

3.19 Circuit identification.

3.19.1 Electrical/electronics circuits. Identification of electrical/electronic circuits within the module shall be in accordance with MIL-STD-681, system II, except that miscellaneous circuits shall be coded in accordance with system I.

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3.19.2 Identification of interconnecting wire. Interconnecting wire shall be identified by means of marking on white/insulated wires or applying sleeves, tags or pressure-sensitive tapes on the insulated wires.

3.20 Motors.

3.20.1 General. Motors shall be in accordance with MIL-M-17060 and the following unless otherwise specified in the detail specification:

Classification	Service A
Ambient temperature	40°C
Volt	440
Phase	3
Hertz	60
Duty	As specified for winch
Enclosure	Watertight
Speed	Constant
Type	Squirrel cage induction
Design	Type B
Horsepower	As specified
Insulation	Class H

3.20.2 Sizing. Where required horsepower value falls between standard motor sizes in accordance with MIL-M-17060, the next higher standard rating shall be used.

3.20.3 Rotation. All motors used shall be marked in a permanent manner to show direction of rotation where applicable to the function of the equipment.

3.20.4 Identification plates. All motors shall be fitted with identification plates of corrosion-resistant material marked with the following information:

Manufacturer's type and frame designation
Output horsepower (hp)
Kind of rating
Temperature rise at rated load and design ambient temperature
Rpm at rated load
Volt, phase, frequency
Amperes at rated output

3.21 Motor controllers and master switches.

3.21.1 General. Motor controllers and master switches shall be in accordance with MIL-C-2212 or MIL-C-23742, as applicable and as follows, unless otherwsie specified in the detail specification:

Operation	Magnetic
Volt	440 vac, 3 phase, 60 Hz
Type	Across-the-line
Performance	Nonautomatic
Function	Motor and electronics starting, stopping
Protection	Low volt protection and thermal overload
Enclosure	Dripproof
Duty	Continuous
Emergency run feature	Remote emergency run pushbuttons shall be provided with a watertight housing
Ambient temperature	50°C
Indicating lights	Green indicating light shall be provided on the winch control panel to indicate the motor running

Master switch:

Volt	120 volts
Location and type	Pushbutton, mounted or built into the winch control stand
Enclosure	Watertight with space heaters in accordance with MIL-C-2212

3.21.1.1 No control circuit voltage within the master control panel shall exceed 120 volts.

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3.21.2 Marking. Motor controllers and master switches shall be legibly and permanently marked as to the function of each position.

3.22 Electro-mechanical.

3.22.1 General. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence of the same. The design layout and assembly of the units and their component parts shall be such as to facilitate quantity production and to result in minimum size and weight.

3.22.2 Parts. Parts shall be selected as specified herein. The fact that standard parts, as required herein, are used in the equipment does not relieve the supplier of the responsibility for conforming with all performance requirements specified in the individual equipment specification for the particular equipment.

3.22.2.1 Combining electronic and mechanical devices within the same assembly shall be avoided. Where the design requires electro-mechanical assemblies, electronic parts shall be mounted or grouped to permit easy removal and decoupling from the mechanical portion of the assembly.

3.22.3 Selection of parts. All parts used in the construction of electronic equipment shall be in accordance with the following:

(a) Military specifications.
(b) MS parts.
(c) MIL-STD-242.
(d) Other standards, specifications, and requirements specified herein but not included in (a) through (c) above.

3.22.3.1 Derating of parts. Component parts such as resistors, capacitors, etc., subjected to internal or external thermal changes shall be properly derated so that a continuous duty cycle can be guaranteed.

3.22.4 Heat dissipation. Where derating is not sufficient to protect the operation of the unit, heat removal methods shall be employed.

3.22.4.1 Heat removal method. The following criteria shall determine the applicability for cooling:

(a) Total heat dissipation exceeding 5000 watts.
(b) Unit heat dissipation exceeding 10 watts per square foot of enclosure surface area.

3.22.4.2 Equipment which does not meet the criteria of 3.22.4.1, equipment intended for shipboard use, and portable equipment shall be air cooled, using compartment (room) air as the heat sink.

3.22.5 Prohibited materials. Materials in this category shall be prohibited as in MIL-E-917. Examples of the type of material specified are as follows:

(a) Toxic materials.
(b) Flammable materials.
(c) Fragile materials.
(d) Mercury.
(e) Magnesium.
(f) Radioactive materials.

3.22.6 Metals. Metals shall be selected or processed and applied in a manner that provides corrosion-resistance. Metals that are not inherently corrosion-resistant shall be processed (treated, plated, or painted) to provide corrosion-resistance.

3.22.6.1 Selection of metals in direct contact. In order to minimize corrosion attack due to electrolytic action between dissimilar metals, in contact with each other, metal-to-metal contacts shall be limited to those metals which, when coupled, are designated by an open squre, an open delta, or an open delta with a cross bar MIL-E-917 under the following conditions:

(a) For combinations of metals subject to immersion, splashing, or spray from seawater, condition S, E, or L, as applicable shall apply.

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(b) For combinations of metals exposed to atmosphere but not subject to immersion, splashing or spray from seawater, condition D shall apply.

If a metal is coated or plated, the coating or plating metal rather than the base metal shall be considered. Metal-to-metal contact is considered to exist between parts that depend upon painting for corrosion-resistance.

3.22.7 Nonmetals.

3.22.7.1 Plastics. Plastics which serve as electrical insualtion shall be in accordance with MIL-E-917. Plastics which do not serve as electrical insulation (structural parts, etc) shall meet all physical and mechanical properties required for plastic insulating materials, including nonflammability and nontoxicity; however, these plastics need not meet the arcing and tracking resistance requirements of MIL-E-917.

3.22.8 Encapsulating materials. Modules shall be encapsulated by the following:

(a) Use of rigid materials (i.e., vacuumed or thermetically sealed receptacles in accordance with MIL-I-16923.
(b) Impregnations, filling, potting encapsulating compounds. The compound selected shall not flow at temperatures encountered during the temperature operating tests, unless contained in such a manner that the compound will not flow from the container.

3.22.9 Fungus - inert material. Equipment or components shall be fungus-proofed by selection of parts and materials that are now nutrient for fungus, or the parts and materials shall be so treated prior to their use in the equipment that overall spraying of the equipment is not necessary.

3.22.10 Protection of aluminum and aluminum alloys. Unpainted aluminum and aluminum alloy parts shall be anodized in accordance with MIL-A-8625. Where anodizing interferes with electrical properties, chemical treatment conforming to MIL-C-5541 may be used. Prior to painting, aluminum and aluminum alloys shall be anodized in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541.

3.22.10.1 Cadmium. Cadmium plating shall conform to type II, class 1 of QQ-P-416, except for threaded devices (see 3.22.17.4) which shall be type II, class 3 of QQ-P-416.

3.22.11 Fabrication.

3.22.11.1 Boxes, cases, shields, and compartment walls shall be made by casting, drawing, or bending, and welding, except when ease of servicing of the equipment requires that a removable panel construction be used, or when the applied stresses dictate the use of a strong aluminum alloy which does not prove a good weld; for such parts, riveting or bolting may be used.

3.22.11.2 Tolerances. In the selection of parts, other than those listed in MIL-STD-242 or otherwise specified herein, the widest tolerances permitted by the individual part specification commensurate with the particular long term stable circuit application requirements shall be used. However, in the fabrication of the equipment, it is permissible to substitute parts of closer tolerances as long as interchangeability is not affected.

3.22.11.3 Accessibility. Each article of equipment and each major subassembly forming a part thereof, shall provide easy and ready access to its interior parts, terminals, and wiring for adjustments, complete circuit checking, and the removal and replacement of component parts.

3.22.11.4 Connections to parts. Connections to parts inside a removable container shall be arranged to permit the removal of the container without threading connection leads through the sides or top of the container.

3.22.11.5 Parts which are identified as replaceable by repair parts for the equipment shall be easily removable.

3.22.11.6 Electrical bonding. Protective finishes shall be omitted at those points where their presence would prevent proper electrical bonding as required for shielding or connection. Provision shall be made to assure permanence of electrical contact between the surfaces of all parts in contact over long periods of time or in the presence of humid saline atmospheres.

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3.22.11.7 Cabinet bonding. All equipment cabinets shall be furnished with mounting holes or studs for bonding strap ground connector. The holes or studs for bonding strap connector shall not reduce the spraytight, dripproof, or watertight characteristic. The holes or studs shall not interfere with the extraction or insertion of the equipment into the cabinet or the placement of the cabinet in a rack, or shock mounting, if required.

3.22.11.8 Gaskets. Wherever practicable, gasketed surfaces shall have metal-to-metal contact, so that the amount of gasket compression can be accurately controlled. The inside radius of gaskets at corners of square or rectangular enclosures shall be the maximum value that will not restrict access or inhibit design. The minimum radius shall be consistent with the gasket material.

3.22.12 Connections.

3.22.12.1 Mechanical connections. All connections shall be so supported as to obviate breakage and minimize changes in performance due to vibration, inclination, or shock encountered under the specified service conditions.

3.22.12.2 Clamp contact connections. In no cases shall electrical connections depend upon wires, lugs, terminals and the like clamped between a metallic member and an insulating pliable material. Such connections shall be clamped between metal members.

3.22.12.3 Power receptacles. Convenience receptacles shall not be provided.

3.22.12.4 Connectors, externally mounted. All connectors mounted on the external surface of the equipment shall be in accordance with MIL-C-22992, or MIL-C-26482 (bayonet coupling). Connector plugs or receptacles which continue to be energized after unmating shall have socket contacts.

3.22.12.5 Connectors, general. The mating connectors shall be furnished with all receptacles or plugs. All external mounted connector receptacles shall be provided with a suitable protective cap to prevent damage to the connector when the mating connector is not installed. The protective cap shall be affixed adjacent to the connector receptacle. The backend accessory hardware necessary to provide the moisture seal to single jacketed cable shall be furnished with each mating external connector plug.

3.22.12.6 Connectors, removable contact, crimp type. Environmental connectors shall have the full complement of contacts installed.

3.22.12.7 Terminal boards, terminal strips, and connectors. Terminal boards, terminal strips and multicontact connectors used in interconnecting major assemblies and multicontact connectors used in interconnecting major units shall have 10 percent extra unused terminals or contacts. No spare contacts are required on multicontact connectors used for primary power connections. Stud type terminal boards shall be in accordance with MS27212.

3.22.12.8 Banana plugs. Banana plugs shall not be used.

3.22.12.9 Soldering connections. Pigtails (wire leads which are permanently affixed to a part), studs (soldering posts which are permanently affixed to a part), soldering lugs (projections which solder to end of wires to facilitate attachments to solderless connections), and soldering terminals (items which are attached to the chassis, or insulating boards, strips, or posts to serve as a solder junction for two or more wires or leads) come under this heading. All soldering studs, lugs, and terminals shall be notched or otherwise provided with means for mechanically securing the wire or lead prior to the application of solder. All solder type studs and terminals shall be sturdily designed and shall be constructed of materials which will render them suitable for repeated soldering and unsoldering operations without breaking or loosening. All solder type studs shall be mounted in such manner as to preclude their loosening or rotation due to soldering operations or from strains due to attached wires or leads. No more than three wires, including wires from parts, shall terminate at any one terminal, except for double turret terminals which may terminate three wires per section.

3.22.12.10 Wire and connections. The ends of each wire (except for pigtail leads or parts requiring solder connections) shall be terminated by solderless crimp type lugs, subject to the following:

(a) Insulated, solderless lugs are preferred and shall conform to MIL-T-7928.
(b) Where thermal or other considerations prevent the use of insulated lugs noninsulated solderless lugs conforming to MIL-E-16366 shall be used, provided they are covered with an insulating sleeve.

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3.22.12.11 Wire size. Minimum standard wire size permitted shall be AWG 18 for interconnecting wire and AWG 20 for chassis wiring. Hookup wire shall conform to MIL-W-16878 type E and interconnecting wire shall conform to MIL-W-16878, type EE.

3.22.12.11.1 Insulation protection. Wherever wires are run through holes in metal partitions (shields and the like) less than 1/8 inch in thickness, the holes shall be equipped with grommets for mechanical protection of insulation otherwise subject to abrasion. Panels 1/8 inch or more in thickness shall have either grommets or the hole edges rounded to a radius equal to one-half the thickness of the material. Grommets for wires operating at radio frequency (rf) potentials shall be of ceramic, styrene, or phenolic, except for coaxial cables which have outside protection, where rubber or neoprene is acceptable. Care shall be exercised in the running of wires to insure that they are not carried over or bent around any sharp corner or edge which might in time cut through the insulation.

3.22.13 Controls, indicators, and panel layout.

3.22.13.1 The number of controls and indicators shall be kept to a minimum consistent with the requirements of the individual equipment specification. Clockwise rotation of control knobs and handles shall increase the final controlled effect, except where mechanical or other complications make it impractical.

3.22.13.2 Position identification. Operating controls whose relative position give an indication of equipment performance shall be marked, indexed, or metered in such a manner that the control position can be readily identified.

3.22.13.3 Arrangement. Controls shall be so arranged as to facilitate smooth and rapid manipulation. Indicators shall be designed and arranged to insure readability under service conditions. Location of similar controls and indicators on different panels shall correspond insofar as practicable.

3.22.13.4 Illumination. Controls, switches, and dials shall be adequately illuminated, either internally or externally, so that they may be read easily and without undue fatigue. All lights shall be dual type, low voltage (28 volts), and be controlled by a single dimmer except for warning lights which shall not be controlled by a dimmer. The dimmer shall control the lights from zero to full brilliance. The lights shall provide uniform illumination without glare or stray light. Where the observation of an object or surface is critical to the operation of equipment, the illumination shall be from two or more sources. Pilot lights shall not be connected in series.

3.22.14 Enclosures.

3.22.14.1 Electronic control box. The electronic control box shall be dripproof protected in accordance with MIL-STD-108 enclosures for motors.

3.22.14.2 Master station control panel. The master station control panel shall be watertight in accordance with MIL-STD-108 and shall be provided with suitable 115 volt heaters designed for continuous operation from the ship's lighting circuit.

3.22.14.3 Motor controller and master switch. Enclosure requirements for the motor controller and master switch shall be as specified in 3.21.1.

3.22.14.4 Motor. Enclosure requirements for the motor shall be as specified in 3.20.1.

3.22.14.5 Remote station. Remote station shall be watertight in accordance with MIL-STD-108.

3.22.14.6 Test receptacles. Test receptacles shall be environmentally protected dependent upon their location.

3.22.15 Components.

3.22.15.1 Transformers. Standard transformers in accordance with MIL-STD-242 shall be used to the maximum extent possible. Where transformers, inductors and chokes are not available on the Qualified Products List of MIL-T-27 and the number of identical units per contract or order is less than 100, it will not be necessary to qualify these items for inclusion on the Qualified Products List of MIL-T-27.

3.22.15.2 Semiconductor. Semiconductor devices shall be selected from MIL-STD-701 and shall conform to MIL-S-19500 with respect to performance. Germanium diodes and germanium transistors shall not be used.

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3.22.15.3 Clamps. All plug-in items shall be secured with positive holding clamps and shall be capable of being easily released for replacement. These clamps shall be sufficiently sturdy to retain the parts in their proper position under the specified conditions of shock and vibration.

3.22.15.4 Fuses. Fuses shall conform to MIL-F-15160. All fuses shall be easily replaceable. On extra fuse of each type and ratin used shall be supplied and attached at a convenient location on the equipment for replacement purposes, except for miniature equipment where this may not be feasible.

3.22.15.5 Electrical tubing and sleeving. Nonrigid tubing shall conform to MIL-I-631 or MIL-I-7444, and MIL-L-I-23053. Sleeving shall conform to MIL-I-3190.

3.22.15.6 Tape, lacing. Lacing tape for cable harnesses and other applications shall conform to type P, waxed, class 2 of MIL-T-713.

3.22.15.7 Meters, ruggedized. Indicating meters for use in equipment required to withstand shock shall be ruggedized in accordance with MIL-M-10304 and shall be in accordance with the following:

(a) Adjustment. All panel indicating meters shall have external zero adjusters.
(b) Iron and direct current fields. Indicating meters shall be located within the equipment in such a manner that neither large masses of iron nor direct current fields will affect the accurate operation of the meters. Iron vane type meters shall be so located that transformers or choke coils carrying current will not affect the accurate operation of the meters.
(c) Shielding. In the application of external thermocouple rf meters especially for currents above 1 ampere, care shall be taken to provide adequate shielding of both the meter and leads so that stray rf current does not flow back through the meter leads to the thermocouple. Care shall also be taken with the meters operating at high rf voltage above ground to ensure that stray capacity currents through the meters to the metal panel of the equipment are not appreciable enough to cause errors or damage in the meter.

3.22.15.8 Shunts. External instrument shunts shall conform to MIL-S-61.

3.22.15.9 Relays. Armature type relays shall conform to MIL-R-5757 or MIL-R-6106. All other types of relays shall be hermetically sealed with solder terminations. Thermal time delay relays shall conform to MIL-R-19648. Relays containing mercury in any form shall not be used.

3.22.16 Switches.

3.22.16.1 Rotary switches. Rotary switches shall be as follows:

(a) High current. High current switches shall conform to MIL-S-15291 (snap action) or MIL-S-21604 (selector). All switches shall be of the positive action type.
(b) Low current. Low current (up to 2 ampere) switches shall conform to MIL-S-3786.

3.22.16.2 Switch assemblies and actuators; pushbutton and limit. Switch assemblies and actuators shall be in accordance with MIL-S-8805.

3.22.16.3 Pushbutton and limit. Pushbutton and limit switches shall be in accordance with MIL-S-8805. Limit switches shall be actuated indirectly by lever arm or other means.

3.22.16.4 Toggle. Toggle switches shall comply with MIL-S-3950 or MIL-S-8834.

3.22.16.5 Pushbutton, lighted. Lighted pushbutton switches shall be in accordance with MIL-S-22885.

3.22.16.6 Coaxial. Coaxial switches shall be selected from MIL-HDBK-216 and shall conform to the latest issue of the specifications and drawings referenced therein.

3.22.16.7 Thermal. Thermal switches shall have a bi-metal sensing element and shall be factory calibrated and sealed for temperature settings. The opening and closing of contacts as to temperature shall be legibly marked on the switch or frame, as applicable.

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3.22.17 Fasteners. Fasteners, threaded devices, threaded bolts, nuts, screws and studs employed in the construction of equipment shall conform to FF-S-85, FF-S-86, FF-S-92, FF-S-200, FF-S-107, MIL-B-857, MIL-N-25027 and FF-S-210.

3.22.17.1 Screw threads. Screw threads for all threaded securing devices shall be of the unified thread series in accordance with Handbook H28 and supplement thereto and shall be of the coarse or fine thread series as specified herein.

3.22.17.2 Material. All screws, studs, and spade bolts and nuts shall be of an austenitic corrosion-resistant steel, a nickel-copper alloy or cold rolled carbon steel material cadmium plated in accordance with type II, class 3 of QQ-P-416, except in those cases where the magnetic or conductivity properties of the above materials are detrimental to the operation of the equipment. In such cases, brass or a nonferrous alloy of comparable strength and conductivity which is protected against corrosion shall be employed. No aluminum nuts shall be used. These requirements do not apply to parts or assemblies wherein the use of threaded devices is governed by the part or assembly specifications.

3.22.17.3 Thread engagement. Screws and bolts in tapped parts other than nuts shall enter and have full thread engagement for an axial distance at least equal to the diameter of the externally threaded fastener without bottoming.

3.22.17.4 Thread projection. Screws, secured by nuts, or other retaining devices which permit projection beyond the retaining device shall be of such length to permit a minimum projection of 1-1/2 threads and a maximum projection of 1/8 inch plus 1-1/2 threads for screws up to and including 1 inch in length and 1/4 inch plus 1-1/2 threads for screws over 1 inch in length. These requirements shall not be construed as precluding the use of screws assembled in blind tapped holes in castings, or spacers, or where the design restrictions require that the screw length be such as to require that the threaded portion of the screw be flush with the retaining device.

3.22.17.5 Nonmetals. Where screw threads engage tapped holes in nonmetals, such tapped holes shall be provided with metallic inserts secured by knurling or other method which will preclude their movement and is acceptable to the Government.

3.22.17.6 Panel and cover plate retaining devices. Screws or similar devices employed for retaining front panels or removable cover plates shall be of captive types, and shall employ slotted heads or knurled and slotted heads. They shall be of such design as to be capable of definite individual loosening prior to movement of the part to which they are captive. Design and construction shall be such as to provide for their self-alinement with their retaining nuts, blocks, or inserts without sticking and without damage to their threads. Since the body of a captive screw is often reduced to less than the root diameter of the threads, the screw acts only in tension and carries no shear load. For heavy chassis, sufficient friction is seldom present between the front panel and the frame to prevent movement during vibration and shock conditions. In order to eliminate this possible source of trouble, the use of a combination captive screw and guide pin is encouraged. Satisfactory assemblies incorporating this feature are commercially available. Quarter-turn fasteners may be used as panel, other than front panel, and cover plate retaining devices for nonstructural applications and shall conform to MIL-F-5591. (Panels and chassis on drawer slides are considered structural applications.)

3.22.17.7 Nuts. Hexagonal nuts are preferred for general usage. Square nuts may be used only when they are captive or floating as part of a fastening device.

3.22.17.8 Nuts, self-locking. Self-locking nuts shall conform to MIL-N-25027. (Where required, nuts for screws larger than number 8 or the coarse thread series may be utilized.) When nonmetallic inserts are employed as the locking medium, only nylon (polyamide) shall be used. The use of self-locking nuts with nylon inserts shall be limited to a maximum temperature of 250°F. Self-locking nuts of the fibre binding type shall not be used. For temperatures in excess of 250°F, all metal self-locking nuts shall be used. Self-locking nuts shall not be used for ground connections.

3.22.17.9 Panel-mounting screws. Panel-mounting screws shall be limited to the pan-head or flat cross recessed-head screws. Round-head screws shall not be used for panel mounting.

3.22.17.10 Rack-mounted panels. Rack-mounted panels shall be held by oval head screws with cup type washers.

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3.22.17.11 Application. All nuts, other than self-locking nuts, and machine screws shall require a lockwasher, or equivalent, under the nut. Where lockwashers are used externally for securing to an aluminum part, a flat washer shall be provided to prevent marring the protective coating. All lockwashers and flat washers shall be fabricated of corrosion-resistant metals or shall be so treated as to resist the corrosive effects of a moist sea atmosphere.

3.22.17.12 Countersinking. Screws should not normally be countersunk in sheet material having a thickness less than one and one-half times the height of the screw head (in no case shall a feather edge be created). Thinner material should be dimpled to assure proper seating, with the screw head edge flush or below the surface of the material.

3.22.17.13 Adjusting devices. All threaded securing devices used for adjustment purposes shall conform to the unified coarse or fine thread series.

3.22.17.14 Locking devices. All screw thread assemblies shall be made vibration-proof so that no loosening results under the vibration and shock requirements specified in the contract.

3.22.17.15 Control shafts and couplings. Wherever the proper functioning of the device is dependent upon maintaining a prescribed relationship between control shafts and associated couplings, collars, gears, and similar parts not permanently fastened to the shaft, such parts shall be secured by positive locking devices. Acceptable methods of attachment are keys, splines, pins, and suitable flat or rectangular shapes which shall be provided on both of the connecting parts. Light-duty items, such as indicator mechanisms and other devices normally designed to use friction locking means for adjustment, are exempted from this requirement.

3.23 Control system.

3.23.1 Servo control. An electro servo control system shall be provided for controlling the winch. The control shall consist of a motor controller, electronic control center, a master station operator controlled panel (master switch for motor start/stop may be integrally mounted on this panel), and an integrally mounted winch control. Other controls shall be as required.

3.23.1.1 Environmental parameters shall be as described in this specification.

3.23.2 All winches shall contain a minimum of two speed control units as follows:

(a) A speed control unit in the master station.
(b) A speed control unit mounted on the winch.

3.23.3 Master speed control unit.

3.23.3.1 Master speed control unit shall be mounted on the master control panel booth.

3.23.3.2 The control shall contain a lever which shall be capable of moving through an arc of approximately 120 degrees for movement of the hydraulic pump from full stroke in the hoist direction to full stroke in the payout direction. Pulling the lever shall cause the pump to go on stroke in the hoist (haul in) direction and pushing the lever shall cause the pump to go on stroke in the lowering (payout) direction.

3.23.3.3 The control system shall be designed to withstand a force of 200 pounds when applied to the control lever.

3.23.3.4 The control handle should not exceed 10 inches and shall be self-centering.

3.23.3.5 The effort to operate the control shall not exceed 10 pounds.

3.23.3.6 Speed control unit lever shall be provided with a positive spring return to center (zero stroke) position, the returning time shall not exceed one second.

3.23.4 Winch speed control.

3.23.4.1 An emergency mechanical speed control unit shall be provided as an integral part of the sealed transmission. The control shall provide speed and direction control of the winch by means of a mechanical servo control system. This control is intended for use only in the event the electronic control system is inoperative. The maximum speed obtainable from this control shall be limited to the rated speed specified for the winch.

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3.23.4.2 A box wrench shall be used for stroking the pump through the mechanical speed control unit. The control handle (box wrench) shall move through an arc of approximately 120 degrees for movement of the pump from full storke in the hoist direction to full stroke in the payout direction. The control system shall be designed to withstand a force of 200 pounds when applied to the control handle. The effort to stroke the pump shall not exceed 150 inch-pounds.

3.23.5 Secondary (remote) control units.

3.23.5.1 Where one or more secondary (remote) speed control units may be required, the control units shall be mounted away from the winch or master station and as required by the ship's specification or contract or order.

3.23.5.2 The control units shall be similar in appearance and operation to the control unit described in 3.23.3.

3.23.6 Where additional types of control units are required, such as tension and position controls, appearance and operation shall be as described in the equipment specification.

3.23.7 All control operations shall be selected from the master control panel located in the control booth.

3.23.7.1 Standard nomenclature. Operations shall be selected through the use of a rotary selector switch.

3.23.7.2 Positions. Positions shall be as follows:

(a) Master speed shall be in the center top (12 o'clock) position above the selector pointer knob.
(b) Winch speed control shall be oriented to the right (clockwise movement of the selector knob), of the master speed position.
(c) Secondary (remote) speed control, when required, shall be to the right, (clockwise movement of the selector knob), of the winch speed control.
(d) Where more than one secondary speed control is required, the actuation position shall be further to the right of the preceeding secondary speed control.
(e) Where a master position control is required, it shall be located to the left (counterclockwise movement of the selector knob), of master speed control.
(f) Where a tension control is required, it shall be located further to the left, (counterclockwise movement of the selector knob), of the master position control.
(g) All positions shall be firm. Where an operation is not required, that spot on the selector shall be left blank.

3.23.8 Fail-safe.

3.23.8.1 The electronic control system shall incorporate a fail-safe system which will automatically return the hydraulic pump to zero stroke uder the following conditions:

(a) Any malfunction of an electronic, hydraulic or mechanical component, which would result in a runaway payout or hand-in condition at the winch drum.
(b) Any malfunction of an electronic, hydraulic or mechanical component, which would result in the loss of winch capability to maintain proper tension in the rope during tension mode operations or loss of winch capability to support a load during speed mode operations.
(c) Loss of any stage or portion of the electronics reference power supply or the loss of 440 volt or 115 volt power supplies.

3.23.8.1.1 Whenever the electrical fail-safe is actuated a red warning light located on the winch control power shall be illuminated.

3.23.8.2 Response. All control systems shall be designed to electronically limit the rate at which an operator signal, whether accidental or intentional, is accepted. This feature shall limit excessive response or higher transient loads to the winch system.

3.23.8.3 Electrical interlocks. Electrical interlocks shall not be provided in the control systems to prevent starting of the electrical motor except when hydraulic control is in the neutral position. However, electrical interlocks shall be provided to prevent starting of the electric motor except when the selector switch is in speed control position.

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3.24 Winch control. The winch control shall be designed so that the hydraulic brake will be released at not more than 4 percent pump stroke and shall set at not less than 3 percent stroke in either direction. The hydraulic brake control valve shall be operated direct or through a pilot valve shall be operated direct or through a pilot valve as required to provide the accuracy specified herein. Cross venting or open center shall be provided between the high pressure and low pressure hydraulic line when the hydraulic brake is set.

3.25 Performance.

3.25.1 The winch shall be designed for the line pull and line speed specified herein. Rating of the winch shall be with the wire rope on the average layer.

3.25.2 The winch shall provide a light line speed, hoisting and lowering, of not less than 180 percent rated full load hoisting speed.

3.25.3 The winch shall be capable of hoisting 150 percent rated load with the wire on the average layer.

3.25.4 With the electric motor energized and the winch control in neutral the winch shall be capable of stopping any load to a maximum of 150 percent rated load without drifting The stopping and holding of the load shall be by setting of the hydraulic brake.

3.26 Painting.

3.26.1 Before assembly all ferrous surfaces not intended to be bright shall be cleaned by abrasive blasting in accordance with the SSPC No. 10 and shall be coated as follows:

(a) One coat of inorganic zinc silicate coating selected from Chapter 9190 of NAVSHIPS 0901-000-0100, paragraph 9190.111.
(b) Over the inorganic zinc primer apply one coat at 1.5 mils (minimum dry film thickness) of zinc yellow-iron oxide in accordance with TT-P-57, type II.
(c) After assembly all primed surfaces shall be top coated with two coats of enamel in accordance with TT-E-490 (color-Navy haze gray). Total top coat dry film thickness shall be not less than 5 mils.

3.26.2 Before assembly all nonferrous surfaces shall be primed with formula 117 (approximately 0.5 mils). After assembly all primed surfaces shall be top coated with two coats of enamel in accordance with TT-E-490 (color-Navy haze gray). Total top coat film thickness shall be not less than 5 mils.

3.26.3 Surfaces to be painted shall be cleaned free of all grease, oil, dirt, and other contaminants, prior to painting. Solvents or detergent washes shall be used to remove such contaminants. Finally, the surface shall be wiped dry, or flushed clean with fresh water if appropriate, and dried before painting.

3.27 Identification plates. Identification plates shall be made from either brass, nickel-copper alloy, or corrosion-resisting steel and shall be a minimum thickness of 0.03 inch. Stamping of letters, numerals, and other characters shall be not less than 0.003 inch deep. Identification plates shall include type of winch, National stock number, manufacturer's identification, contract or order number, serial number, date of manufacture, and weight of winch.

3.27.1 A complete winch electrical schematic diagram identifying terminal numbers and designators shall be legibly reproduced on plastic and mounted on the inside of the electronics cabinet door.

3.28 Onboard repair parts. One set of repair parts shall be provided for one to five winches, of the same type and rating, installed on each ship. Two sets shall be provided for six to twenty units and three sets for twenty-one or more winches.

3.28.1 Mechanical. Each set of mechanical repair parts shall consist of the following items as applicable, in sufficient quantity for complete replacement for one winch:

(a) One set of bushing type bearings.
(b) One set of antifriction bearings.
(c) One set of seals and special packing.
(d) One set of brake linings and rivets.
(e) One set of brake separator disks.
(f) One complete set of springs.
(g) One complete set of mechanical repair parts for the hydraulic pumps and motors in accordance with MIL-P-17869.

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3.28.2 Electrical. Electrical repair parts shall be furnished in accordance with MIL-M-17060 and MIL-C-2212. Electronic parts required to insure one year operation, shall be listed by the supplier and submitted to NAVSEC for review. Electronic parts shall be supplied in the same manner as electrical repair parts are supplied.

3.29 Technical data. The supplier shall prepare engineering drawings, technical manuals and calculations in accordance with the data ordering documents included in the contract or order (see 6.1.2).

3.29.1 Drawings. In addition to the drawing content required by the data ordering document, the unique technical features specified in 3.29.1.1 through 3.29.1.3 shall be included.

3.29.1.1 General arrangement drawings. General arrangement drawings shall contain the following information in addition to the assembly views for the purposes of coordination:

(a) List of electrical, mechanical and hydraulic parts and assemblies.
(b) Duty ratings.
(c) Bedplate or foundation drawings.
(d) An index of all related drawings.
(e) Calculated and actual weight and center of gravity shall be indicated. Also weight of major components shall be listed.

3.29.1.2 Preliminary drawings. Preliminary drawings submitted for review shall consist of the following:

(a) General arrangement drawings.
(b) Assembly and subassembly drawings.
(c) Electric motor and controller drawings, including motors and controllers.
(d) Electric schematic drawings.
(e) Hydraulic pump and motor drawings.
(f) Hydraulic circuit diagrams. The hydraulic circuit diagram shall indicate operating pressures, flow direction rates, valve settings and a complete description of operation. The hydraulic circuit shall be certified by a responsible official of the company manufacturing or designing the hydraulics to be an approved design for the operating conditions specified for the winch.
(g) Electronic control circuit diagram. The electronic circuit diagram shall identify all components of the electronic control system. The electronic control circuit shall be certified by a responsible official of the company manufacturing or designing the electronic control system to be an approved design for the operating conditions specified for the winch.
(h) Sufficient detail drawings to enable evaluation of major end items.
(i) The test points of 3.18.2.1(b) and (c) must be cross-referenced between the electronic control circuit diagrams and the mechanical and hydraulic circuit diagrams and schematics.

3.29.1.3 Final drawings. Final drawings shall consist of the drawings specified in 3.29.1.2 and complete detail drawings on all components.

3.29.2 Technical manuals. In addition to the general requirements covered by the data ordering document, the following unique technical features shall be included:

(a) The manual shall be written to the level of a Machinist Mate, 3rd class or an Electrician's Mate, 3rd class. Two preliminary copies shall be submitted to the command or agency concerned for review and for the assignment of identification numbers. These preliminary manuals shall contain a print of all of the drawings listed below:
  (1) General arrangement drawing.
  (2) Assembly and subassembly drawings in sufficient detail to adequately identify parts.
  (3) Electric motor and controller drawings, integrated into electrical/electronic system schematic and wiring diagrams.
  (4) Hydraulic pump and motor drawings.
  (5) Hydraulic circuit diagram.
(b) Separate manuals shall be prepared for the hydraulic transmissions, hydraulic pumps and motors used in the transmission and electronic control. The manual for the winch shall reference the transmission and electronic control manuals and the sealed transmission manual shall reference the pump and motor manuals. The winch manual shall relate the transmission and electronic controls used to the winch.

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(c) The manuals specified in 3.29.2(b) shall contain information necessary for installation, operation, maintenance, repair and parts support (without the services of manufacturer's representatives) to accomplish organizational level (shipboard) maintenance and depot level maintenance. Depot level maintenance items shall be separate or conspiciously identified to set them aside from organizational level maintenance items.

(1) Organizational maintenance. Organizational maintenance (shipboard) is that maintenance which is the responsibility of and performed by a using organization on its assigned equipment. It normally consists of examining, servicing, lubricating, adjusting, and replacement of parts, minor assemblies and subassemblies.

(2) Depot maintenance. Depot maintenance is that maintenance which is the responsibility of, and performed by, designated maintenance activities, to augment stocks of serviceable material, and to support organizational maintenance and intermediate maintenance activities by the use of more extensive shop facilities, equipment, and personnel of higher technical skill than are available at the lower levels of maintenance. It normally consists of repair, modification, alternation, modernization, overhaul, reclamation, or rebuilt parts, assemblies, subassemblies, components, and end items; the emergency manufacture of nonavailable parts; and providing technical assistance of using activities and intermediate maintenance organizations. Depot maintenance is normally accomplished in fixed shops, shipyards, and comparable shore based facilities.

(d) In writing the manuals, excessive use of technical terminology shall be avoided, e.g., "transition" for "change"; "velocity" for "speed"; "coverage" for "meet"; "optimum" for "best"; "comprised of" for "includes"; "activate" for "start"; etc. While the technician probably knows the meaning of these technical terms individually, he has difficulty understanding them when they are crowded together in a sentence. For example, if the technician is given a technical manual with paragraph after paragraph of the following type of sentence he will have a problem comprehending the data:

"The cover is hermetically sealed to preserve the watertight integrity of the motor and of convoluted configuration to enhance its head dissipation characterisitics".

The technician will understand the text if commonly used words are used in place of the technical jargon:

"A sealed power keeps water out of the motor. The cover has fins to help bleed off motor heat".

3.29.3 Calculations. In addition to the general requirements covered by the data ordering document, the unique technical features specified in 3.29.3.1 and 3.29.3.2 shall be included.

3.29.3.1 Stress calculations. A complete set of stress calculations, in duplicate, of gears, bearings, shafts, keys, pawl and ratchet, clutch, holding down bolts, brakes, etc, shall be included with the first submission of drawings. Under rated conditions, the combined stress in no part shall exceed 35 percent of the tensile yield point of the ductile materials used. At no time shall the combined stress in any part of the machine exceed 70 percent of the tensile yield point when the gypsy is subjected to a horizontal pull equal to the breaking strength of a five inch circumference manila rope with a maximum breaking strength of 22,500 pounds, when acting at midlength of the gypsy, through and normal to the axis of the shaft. The combined stress in no part of the machine shall exceed 70 percent of the tensile yield point when the drum is subjected to a pull equal to the breaking strength of the rope on the average layer with the drum pawl engaged or a shock load equal to 300 percent of the rated load with the rope on the top layer. In stress calculations, the strength relationship shown in table I shall exist.

Table I - Strength relationship.

Design strength	Percent of tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160

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3.29.3.2 Heat calculations. A complete set of heat calculations, in duplicate, shall be submitted with the stress calculations. Heat calculations shall show conformance with the requirements specified for the hydraulic system under the duty cycle specified for the winch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination. Each winch shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II. Nonconforming winches shall not be offered for delivery.

Table II - Examination.

Examination	Requirement paragraph
Drum pawl and ratchet, drum brake, drum clutch and hydraulic brake operating and release mechanisms are centrally located at the end of the winch opposite the wire rope lead and clear of gypsy head rope leads.	3.4.1
Drums are guarded to prevent wire rope from jumping the drum.	3.5
The clutch yoke lever can be locked in both the engaged and disengaged positions.	3.6
Each drum is equipped with a drum brake.	3.7
The diameter of the drum brake hand wheel is less than twelve inches.	3.7
Each drum is equipped with a pawl and ratchet.	3.8
Pawl is heavily counter-balanced or spring controlled to fallout of engagement when the load on the pawl is relieved.	3.8
Shaft caps are provided for the ends of the drum shaft not provided with gypsy head(s).	3.9
Each gear case fitted with drain to permit complete draining of gear case.	3.10
Holes for examination of gears provided.	3.10.1
Dipstick provided for measuring oil level in gear case.	3.10.2
Roller or ball bearings used for gear shafts.	3.11
Roller or ball bearings used for drum shafts.	3.11
Needle or ball bearings used for accessories and accessory mechanisms such as drum pawls, clutches, brakes and operating mechanisms.	3.11
Fasteners provided for securing keys, feathers, splines, nuts, screws and collars.	3.13
Guards provided for all rotating shafts with speeds in excess of 100 rpm.	3.14
Pressure grease fittings provided for all bearings.	3.15
Pressure grease fittings provided for sliding clutch (not required if clutch operates in an oil bath).	3.15
Pressure grease fittings are in accordance with MS15004 or MS15005.	3.15
Hydraulic sump tanks are provided with drain located so fluid can be drained from the tank without spilling on the deck.	3.16.4
Drain line is provided with a hand operated valve and end of the drain line is provided with a cap.	3.16.4.1
Hydraulic filters are provided with pressure differential indicators.	3.16.5
If hydraulic transmission uses precharged accumulator it shall be provided with pressure gage for monitoring charge and fittings for recharging the accumulator.	3.16.6
Air vents are provided in all points in the hydraulic system where air will tend to trap.	3.16.7
Pressure gage connections are provided to permit an operating check of the equipment.	3.16.11
Sight gage provided to indicate level of hydraulic fluid within the sump tank.	3.16.1.
An indicator conforming to MIL-I-17244 is installed in the hydraulic sump tank.	3.16.14

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Table II - Examination (continued).

Examination	Requirement paragraph
Hydraulic sump tank indicator is located in an easily monitored position.	3.16.14
The hydraulic transmission is designed so that it can be readily removed from the winch.	3.16.16
An electrical timer is provided as a part of the hydraulic transmission.	3.16.18
The hydraulic brake has an external manual release (hand lever) which will automatically reset the hydraulic brake when the handle is released by the operator.	3.16.19
The effort to release the hydraulic brake does not exceed 60 pounds on the hand lever.	3.16.19
Instruction plate attached to sump tank near fill point which contains instructions on adding fluid to tank.	3.16.20
Electric motors are marked in a permanent manner to show correct direction of rotation.	3.19.3
Identification plates of corrosion resistant material are provided on the electric motors. Identification plates shall show the following:	3.20.4
Manufacturer's type and frame designation Output hp Kind of rating Temperature rise at rated load and design ambient temperature RPM at rated load Volts, phase, frequency Amperes at rated output Identification plate provided.	3.27

4.2.2 Hydraulic cleanliness. All hydraulic components shall be cleaned and examined before assembly. After the hydraulic transmission is assembled the unit shall be checked to insure that the contamination level does not exceed established standards.

4.2.3 In-process examination. Parts of subsystems that will be closed, sealed or covered at final assembly shall be examined for conformance to applicable specifications and workmanship prior to closure.

4.2.4 Tests. Each winch shall be tested for compliance with this specification. Nonconforming winches shall not be offered for delivery. Quality conformance tests shall consist of the tests specified in 4.3.1 through 4.3.7.

4.2.5 Winch controls used for testing. Winch controls used for testing the winch shall be the same control that will be installed onboard ship. Each control shall carry the same serial number as assigned to the winch.

4.3 Test methods.

4.3.1 Static. Before conducting operating tests, each winch shall be statically tested. The winch shall be loaded statically with twice the rated load and held for 10 minutes without damage to or permanent distortion of the equipment. This test shall be conducted with the pawl and ratchet disengaged and drum brake released.

4.3.2 No load. Each winch shall be operated without load at rated load speed for a period of thirty minutes in each direction of rotation. No abnormal heating (150°F) of bearings or operating parts shall occur.

4.3.3 Rated load. Each winch shall be operated, hoisting and lowering rated load at rated speed through a distance of at least 20 feet in each direction for thirty minutes. The load shall be stopped and held suspended for approximately 20 seconds at the end of each cycle. The load should be held suspended at all times.

4.3.4 Overload. Each winch shall demonstrate its ability to hoist and lower 150 percent rated load at no specified speed through a distance of at least 20 feet in each direction for five complete cycles. The load shall be stopped and held suspended for a period of approximately 20 seconds at the end of each cycle. The load should be held suspended at all times.

4.3.5 Drum brake. Each drum brake shall be subjected to a static pull equal to the rating of the brake. No slippage of the drum brake shall occur. The effort to set the drum brake shall not exceed 60 pounds on the handwheel.

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4.3.6 Hydraulic brake.

4.3.6.1 Each hydraulic brake shall be subjected to a static pull equal to 200 percent of the rated load for the winch (wire rope on the average layer) without slippage. Each hydraulic brake shall also demonstrate its ability to stop and hold a load equal to 150 percent of the rated load for the winch when the load is lowered at rated speed under simulated power failure conditions. Stopping distance under this test shall not exceed a distance equal to the distance the winch would payout when running at rated speed for three seconds.

4.3.6.2 Each winch shall be tested to demonstrate that the hydraulic brake will automatically set upon loss of servo pressure or electric power when in speed mode of operation. Each winch shall be tested to demonstrate that the hydraulic brake will stay in the released position for at least fifteen minutes upon loss of servo pressure or electric power when in tension mode of operation.

4.3.7 Heat tests. One out of every ten or fraction thereof of each winch of each type offered for delivery on the contract or order shall be tested to show conformance with requirements of section 3. Necessary adjustments shall be made in the temperature limit specified for the hydraulic fluid when tests are conducted at ambients either higher or lower than the design temperature of 90°F.

4.4 Test procedures and reports. The supplier shall prepare a test procedure and a test report in accordance with the data ordering documents included in the contract or order (see 6.1.2).

4.5 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Hydraulic transmission. The manufacturer of the hydraulic transmission package shall ship the unit fully charged with hydraulic fluid and ready for use.

5.2 Hydraulic seals. To insure that the cleanliness of the hydraulic system is not compromised, seal shall be applied prior to shipment. The seals shall be installed in such a manner that a broken seal would indicate that the system has been compromised.

5.3 Packing. The equipment, accessories, technical manuals and repair parts shall be preserved-packaged level A or C; packed level A, B, or C and marked levels A, B, or C of MIL-P-3184 as specified (see 6.1.1).

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable detail specification.
(c) Level of preservation-packaging, packing, and marking required (see 5.3).

6.1.2 Contract data requirements. When this specification is used a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirement	Service	Applicable DID	Options
(a)	3.29.1	Drawings	SH	UDI-E-23174	Categories E, G and H, form 2 and type II
(b)	3.29.2	Technical manuals	SH	UDI-M-23455	-------
(c)	3.29.3	Calculations	SH	UDI-E-23213	-------
(d)	4.4	Test procedures	SH	UDI-T-23649	-------
(e)	4.4	Test reports	SH	DI-T-2072	-------

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(Copies of DID's required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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